

CMI Entrance Examination

Toppers' Mock Test — Solutions

Part A

Question 1

Answer: ACD

- (a) $f(z) = \sin z$. Setting $\sin z = w$ and rewriting through e^{iz} produces a quadratic in e^{iz} whose roots are never zero. The equation thus has solutions for every $w \in \mathbb{C}$, so $\sin$ is surjective on $\mathbb{C}$.
- (b) $f(z) = e^z$. The exponential omits the value 0 and is therefore not surjective.
- (c) $f(z) = z^2$. Every complex number admits a square root, so the squaring map is surjective.
- (d) $f(z) = ze^z$. This is entire and non-polynomial, so by Picard's Little Theorem it omits at most one value. Since $f(0) = 0$, the value 0 is attained; and for $a \neq 0$, the equation $ze^z = a$ has solutions (the inverse is the Lambert W function). Hence f is surjective.

Question 2

- (a) **True.** By Bertrand's Postulate, for $m > 1$ there is a prime p with $m/2 < p \leq m$. Such a p divides $m!$ exactly once, so $m!$ cannot be a perfect n -th power for any $n > 1$.
- (b) **True.** Take $a = k! - 1$ and $b = k$. Then $a! \cdot b! = (k! - 1)! \cdot k! = (k!)!$, giving infinitely many triples.
- (c) **False.** For every $n \geq 2$, the choice $a = n - 1$, $b = 1$ gives $(n - 1)^1 + 1^{n-1} = n$. The only natural number not so representable is $n = 1$, so the set of unrepresentable values is finite, not infinite.
- (d) **False.** For $n \geq 4$, the sum $1! + 2! + \cdots + n!$ ends in the digit 3. No square or fourth power ends in 3, and modular arguments rule out cubes and higher powers as well.

Question 3

- (a) **Cannot.** Setting $y = 0$ gives $f(x) + f(0) = x^2$, hence $f(x) = x^2 - c$ where $c = f(0)$. Substituting back yields $-2c = xy$ for all x, y , which is impossible.
- (b) **Cannot.** A continuous f with $f(f(x)) = -x$ must be injective and therefore strictly monotonic. But the composition of any monotonic function with itself is strictly increasing, contradicting the strictly decreasing $-x$.
- (c) **Can.** The Weierstrass function is everywhere continuous and nowhere differentiable.
- (d) **Cannot.** The set of continuity points of any function is a G_δ set. By the Baire Category Theorem, $\mathbb{Q}$ is not G_δ in $\mathbb{R}$.

Question 4

Answer: ABCD

- (a) **True.** An odd-degree real polynomial has range $(-\infty, \infty)$ and so attains negative values. Hence $P(x) > 0$ for all x forces even degree.
- (b) **True.** Every strictly positive real polynomial factors into quadratics with non-real conjugate roots, and any such polynomial can be written as a sum of two squares of real polynomials.
- (c) **True.** For an even-degree polynomial to remain positive as $|x| \rightarrow \infty$, the leading coefficient must be positive.
- (d) **True.** An even-degree polynomial with positive leading coefficient tends to $+\infty$ at both ends, so it attains a global minimum.

Question 5

- (a) **False.** Take $f(x) = \arctan x$ or $f(x) = x/(1 + |x|)$: both are continuous and bounded, yet neither attains a global maximum or minimum.
- (b) **True.** A bounded derivative makes f Lipschitz, and Lipschitz functions are uniformly continuous on $\mathbb{R}$.
- (c) **False.** Convergence of $\int_0^\infty f$ does not require $\lim_{x \rightarrow \infty} f(x) = 0$. Tall, thin spikes (or oscillatory examples like Fresnel-type integrals) provide counterexamples.
- (d) **True.** The function $f(x) = x^2$ for $x \in \mathbb{Q}$ and $f(x) = 0$ for $x \notin \mathbb{Q}$ is differentiable exactly at the origin.

Question 6

Answer: ABD

- (a) **True.** Among the 2^{10} subsets, exactly half have even cardinality, giving $2^9 = 512$.
- (b) **True.** Each element independently lies in $A \subsetneq B$, $A = B$, or neither, giving 3^{10} ordered pairs with $A \subseteq B$. Subtracting the 2^{10} pairs with $A = B$ leaves $3^{10} - 2^{10}$.
- (c) **False.** The number of subsets of $\{1, \dots, n\}$ with no two consecutive elements is F_{n+2} . For $n = 10$ this is $F_{12} = 144$, not 128.
- (d) **True.** Each element goes into A , into B , or neither, giving 3^{10} assignments. Removing the cases with $A = \emptyset$ or $B = \emptyset$ (inclusion-exclusion gives $2 \cdot 2^{10} - 1$) and dividing by 2 to unordered the pair yields $(3^{10} - 2^{11} + 1)/2$.

Question 7

- (a) **Cannot.** The identity $\sin^4 x + \cos^4 x = 1 - \frac{1}{2} \sin^2(2x)$ shows the expression is bounded below by $1/2$, so it never equals $1/4$.
- (b) **Can.** In any acute triangle, $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2 \cos A \cos B \cos C < 1$, since all three cosines are positive.
- (c) **Cannot.** The equation reduces to $\sin x + \cos x = \pi/2$, but $\sin x + \cos x \leq \sqrt{2} < \pi/2$.
- (d) **Cannot.** By the Law of Sines, $\sin A + \sin B = \sin C$ translates to $a + b = c$, which violates the triangle inequality for a non-degenerate triangle.

Question 8

- (a) **True.** The interior angles of a convex pentagon sum to 540° , averaging 108° . At least one angle must therefore exceed 100° .
- (b) **False.** The lengths 3, 4, 8 violate the triangle inequality since $3 + 4 < 8$.
- (c) **True.** This is the defining focus–directrix property of a parabola.
- (d) **False.** A regular octagon has interior angle 135° , and 360 is not a multiple of 135, so regular octagons cannot tile the plane on their own.

Question 9

- (a) **True.** This is Nesbitt's inequality.
- (b) **True.** By Cauchy–Schwarz (or the AM–GM inequality applied to each term),

$$\left(\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a}\right)(a + b + c) \geq (a + b + c)^2.$$

- (c) **True.** In a non-degenerate triangle the inequality $a < b + c$ holds; multiplying by a gives $a^2 < ab + ac$, and summing the three cyclic versions yields the result.
- (d) **False.** Take $a = b = 10$, $c = 0.01$: then $abc = 1$ and $ab + bc + ca = 100.2$, while $a + b + c = 20.01$.

Question 10

Answer: BCD

- (a) **False.** The condition $P(A \cup B) = P(A) + P(B)$ forces $P(A \cap B) = 0$, i.e. mutual exclusivity. Mutually exclusive events are independent only in the trivial case where one has probability zero.
- (b) **True.** Independence with $P(A), P(B) \in (0, 1)$ gives $P(A \cap B) = P(A)P(B) > 0$, so the events cannot be mutually exclusive.
- (c) **True.** This is Bonferroni's inequality, which follows from $P(A \cup B) \leq 1$.
- (d) **True.** The implications $P(A | B) > P(A) \iff P(A \cap B) > P(A)P(B) \iff P(B | A) > P(B)$ are equivalent.

Part B

Question 1

- (a) By the Mean Value Theorem, $f'(c_k) = f(k) - f(k-1)$ for some $c_k \in (k-1, k)$.
- For $k = 1$: $f(0) = 0$ and $f(1) \in \{-1, 1\}$, so $f'(c_1) \in \{-1, 1\}$ and $[f'(c_1)]^2 = 1$.
 - For $k = n$: $f(n-1) \in \{-1, 1\}$ and $f(n) = 0$, so $[f'(c_n)]^2 = 1$.
 - For $2 \leq k \leq n-1$: both endpoint values lie in $\{-1, 1\}$, so the difference $f(k) - f(k-1)$ is one of $-2, 0, 2$, giving $[f'(c_k)]^2 \in \{0, 4\}$.

(b) Let $E = \sum_{k=1}^n [f'(c_k)]^2$ and let M be the number of indices k with $2 \leq k \leq n-1$ for which $f(k)$ and $f(k-1)$ differ in sign. Then

$$E = \underbrace{[f'(c_1)]^2}_{=1} + \underbrace{[f'(c_n)]^2}_{=1} + \sum_{k=2}^{n-1} [f'(c_k)]^2 = 2 + 4M,$$

so $E \equiv 2 \pmod{4}$.

Question 2

(a) Apply integration by parts with $u = f(x)$ and $dv = nx^n dx$:

$$n \int_0^1 x^n f(x) dx = \frac{n}{n+1} f(1) - \frac{n}{n+1} \int_0^1 x^{n+1} f'(x) dx.$$

Since f' is continuous on $[0, 1]$, it is bounded, and $\int_0^1 x^{n+1} f'(x) dx \rightarrow 0$ as $n \rightarrow \infty$. Combined with $\frac{n}{n+1} \rightarrow 1$, the limit is $f(1)$.

(b) We compute $L = \lim_{n \rightarrow \infty} n \left(n \int_0^1 x^n f(x) dx - f(1) \right)$. Using the expression from (a),

$$n \int_0^1 x^n f(x) dx - f(1) = -\frac{1}{n+1} f(1) - \frac{n}{n+1} \int_0^1 x^{n+1} f'(x) dx.$$

Multiplying by n :

$$n \left(n \int_0^1 x^n f(x) dx - f(1) \right) = -\frac{n}{n+1} f(1) - \frac{n^2}{n+1} \int_0^1 x^{n+1} f'(x) dx.$$

The first term tends to $-f(1)$. For the second, the same Watson-type lemma applied to f' (i.e. $(n+2) \int_0^1 x^{n+1} f'(x) dx \rightarrow f'(1)$) gives

$$\frac{n^2}{n+1} \int_0^1 x^{n+1} f'(x) dx = \frac{n^2}{(n+1)(n+2)} \cdot (n+2) \int_0^1 x^{n+1} f'(x) dx \rightarrow f'(1).$$

Therefore $L = -f(1) - f'(1)$.

Question 3

(a) By Wilson's Theorem, $(p-1)! \equiv -1 \pmod{p}$. Pair each k in $\{1, 2, \dots, p-1\}$ with $p-k$:

$$(p-1)! = \prod_{k=1}^{(p-1)/2} k(p-k) \equiv \prod_{k=1}^{(p-1)/2} (-k^2) = (-1)^{(p-1)/2} \left[\left(\frac{p-1}{2} \right)! \right]^2 \pmod{p}.$$

Since $p \equiv 1 \pmod{4}$, the exponent $(p-1)/2$ is even, so $(-1)^{(p-1)/2} = 1$. Setting $x = \left(\frac{p-1}{2} \right)!$ gives $x^2 \equiv -1 \pmod{p}$.

(b) Let $N = \lfloor \sqrt{p} \rfloor$, and consider the $(N+1)^2$ pairs (u, v) with $0 \leq u, v \leq N$. Since $N+1 > \sqrt{p}$, we have $(N+1)^2 > p$. Each pair maps to a residue $u - vx \pmod{p}$, of which there are only p . By Pigeonhole, two distinct pairs $(u_1, v_1) \neq (u_2, v_2)$ satisfy

$$u_1 - v_1 x \equiv u_2 - v_2 x \pmod{p}.$$

(c) Set $a = |u_1 - u_2|$ and $b = |v_1 - v_2|$. Rearranging the congruence in (b) gives $u_1 - u_2 \equiv (v_1 - v_2)x \pmod{p}$. Squaring and applying $x^2 \equiv -1$ from (a):

$$(u_1 - u_2)^2 \equiv (v_1 - v_2)^2 x^2 \equiv -(v_1 - v_2)^2 \pmod{p},$$

so $a^2 + b^2 \equiv 0 \pmod{p}$. Since $0 \leq u_i, v_i \leq N < \sqrt{p}$, both $a, b < \sqrt{p}$, hence $0 \leq a^2 + b^2 < 2p$. The pairs are distinct, so a and b are not both zero, giving $a^2 + b^2 > 0$. The only multiple of p in $(0, 2p)$ is p itself, so $a^2 + b^2 = p$.

Question 4

(a) Let $u, v \in \mathbb{R}^2$ be orthogonal. By the Pythagorean theorem,

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2 = \|u - v\|^2,$$

so $\|u + v\| = \|u - v\|$. Since f depends only on the norm, $f(u + v) = f(u - v)$. Substituting into the parallelogram identity,

$$f(u + v) + f(u - v) = 2f(u) + 2f(v) \implies 2f(u + v) = 2f(u) + 2f(v),$$

which gives $f(u + v) = f(u) + f(v)$.

(b) Write $v = v_1 e_1 + v_2 e_2$ in the standard basis. The summands $v_1 e_1$ and $v_2 e_2$ are orthogonal, so by (a) and the radial form $f(x) = g(\|x\|)$,

$$g\left(\sqrt{v_1^2 + v_2^2}\right) = g(|v_1|) + g(|v_2|).$$

Define $h(t) = g(\sqrt{t})$ for $t \geq 0$. Then $h(v_1^2 + v_2^2) = h(v_1^2) + h(v_2^2)$, which is Cauchy's equation on $[0, \infty)$. Continuity of f (and hence of h) forces $h(t) = ct$ for some constant c , so $g(s) = cs^2$ and

$$f(v) = c\|v\|^2.$$

Question 5

(a) The hypothesis is that $p(t)$ is real for real t and non-real for non-real t . Since p takes real values on $\mathbb{R}$, its coefficients are real (a polynomial that is real-valued on an infinite set of reals must have real coefficients). Combined with the non-real-to-non-real condition, we obtain the equivalence

$$p(t) \in \mathbb{R} \iff t \in \mathbb{R}.$$

(b) Fix any real k . If $t \in \mathbb{R}$, then $p(t) \in \mathbb{R}$ by (a), so $p(t) + k \in \mathbb{R}$. Conversely, if $t \notin \mathbb{R}$, then $p(t) \notin \mathbb{R}$, and adding the real number k leaves it non-real. Hence $p(t) + k \in \mathbb{R} \iff t \in \mathbb{R}$.

(c) Let r be a root of $p(t) + k = 0$. Then $p(r) + k = 0 \in \mathbb{R}$, and by (b) this forces $r \in \mathbb{R}$. So every root of $p(t) + k$ is real, for every real k .

(d) Suppose $\deg p \geq 2$. Then p has a turning point, and choosing k to shift the graph of p vertically away from the x -axis produces a quadratic-or-higher polynomial $p(t) + k$ with non-real roots — contradicting (c). So $\deg p \leq 1$. The constant case is excluded since p must send non-reals to non-reals, so $\deg p = 1$ and p is linear.

Question 6

(a) The area of a regular n -gon of side length s is

$$A = \frac{ns^2}{4 \tan(\pi/n)}.$$

If all vertices are lattice points, then $s^2 = (\Delta x)^2 + (\Delta y)^2 \in \mathbb{Z}$, and Pick's theorem forces $A \in \frac{1}{2}\mathbb{Z}$. Rearranging,

$$\tan(\pi/n) = \frac{ns^2}{4A} \in \mathbb{Q}.$$

Niven's theorem states that for rational multiples π/n in $(0, \pi/2]$, the only rational values of $\tan(\pi/n)$ are 0 and 1. For $n \geq 5$, the angle π/n lies strictly between 0 and $\pi/4$, so $\tan(\pi/n) \in (0, 1)$ — which is irrational. No such regular n -gon can exist on the lattice for $n \geq 5$.

(b) For $n = 3$, $\tan(\pi/3) = \sqrt{3}$ is irrational, so no equilateral triangle has all vertices on $\mathbb{Z}^2$.

(c) For $n = 4$, $\tan(\pi/4) = 1$ is rational, and unit squares clearly have lattice vertices. Combined with (a) and (b), the square is the *only* regular polygon all of whose vertices can lie on $\mathbb{Z}^2$.